

06 • Probability, Sampling & Benchmark Statistics

Source: [fanout.sh / inference-eng / 00-04](https://fanout.sh/inference-eng/00-04) · ~8 min

IN SHORT

A model's output is a **random draw** from a probability distribution, so the same prompt gives different text and different lengths. That makes request latency a **skewed distribution with a long tail**: report **P50 / P95 / P99**, not just the mean. Tail percentiles need **many samples**, so always say how many. And a measured difference only counts if it's **bigger than run-to-run noise**.

Two puzzles share one root, randomness: a benchmark gives 41.6 tok/s, then 42.4 tok/s after a settings change (faster?), and asking the model the same question twice gives two different answers.

1. Distributions and sampling

- Given "the sky is", the model outputs a **probability for every possible next token**: blue **62%**, clear **14%**, dark **8%**, everything else **16%**. Each outcome gets a number in $[0, 1]$ and they **sum to exactly 1**.
- Sampling** = one random draw. Lay the probabilities end to end on a 0–1 line, pick a uniform random point; the stretch it lands in is the token.



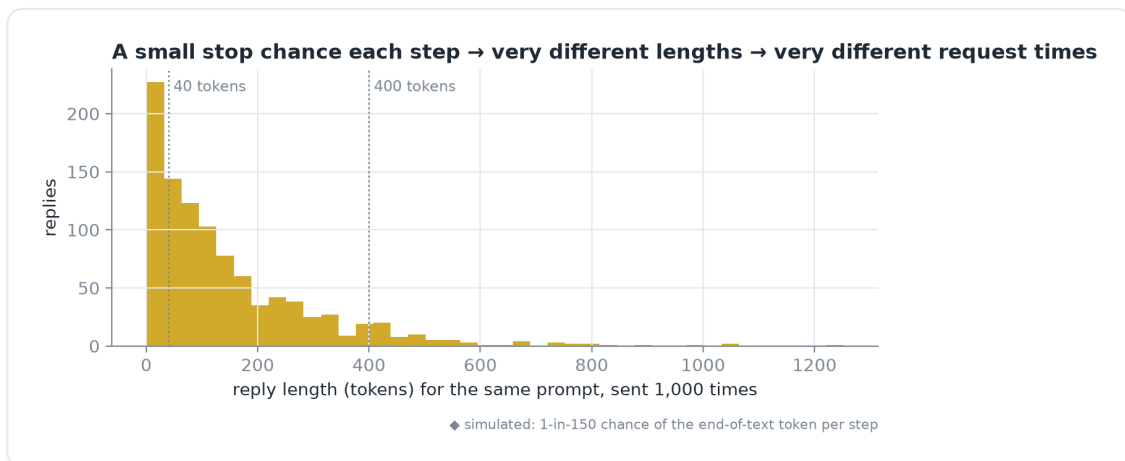
Left: each token owns a stretch of the 0–1 line. Right: over a few draws "blue" wanders; over thousands it settles on 62%.

KEY IDEA

Every token is a fresh draw. That's why the same prompt can give different answers.

2. Randomness leaks into performance

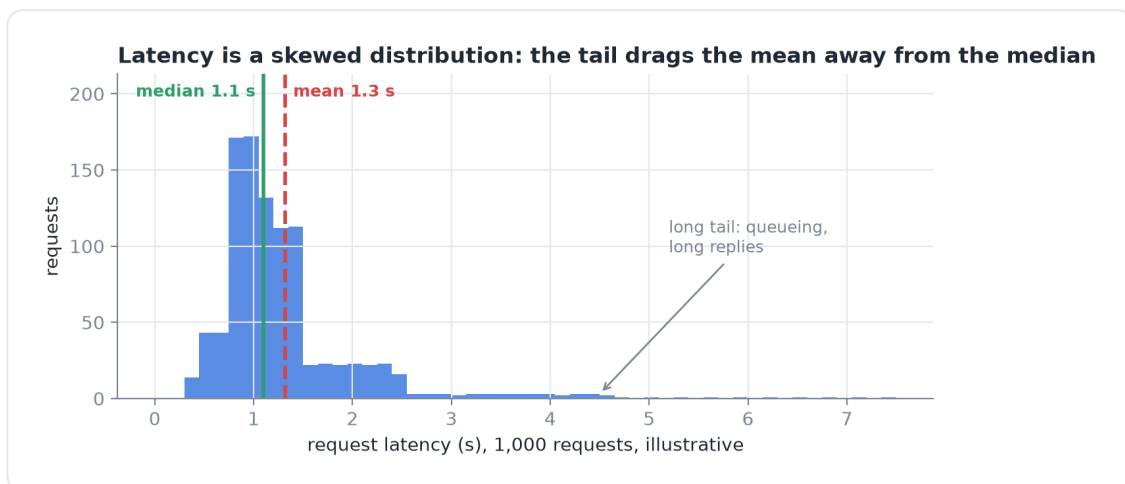
A reply ends when the model draws the **end-of-text token**. A small chance of that at every step means some replies stop after **40 tokens** and others run to **400**. Each decode step costs about the same time, so **reply length decides request time**.



Simulated: the same prompt sent 1,000 times gives a wide spread of reply lengths.

So a benchmark never measures *one* latency; it measures a **distribution** of latencies.

3. Mean, median and the tail



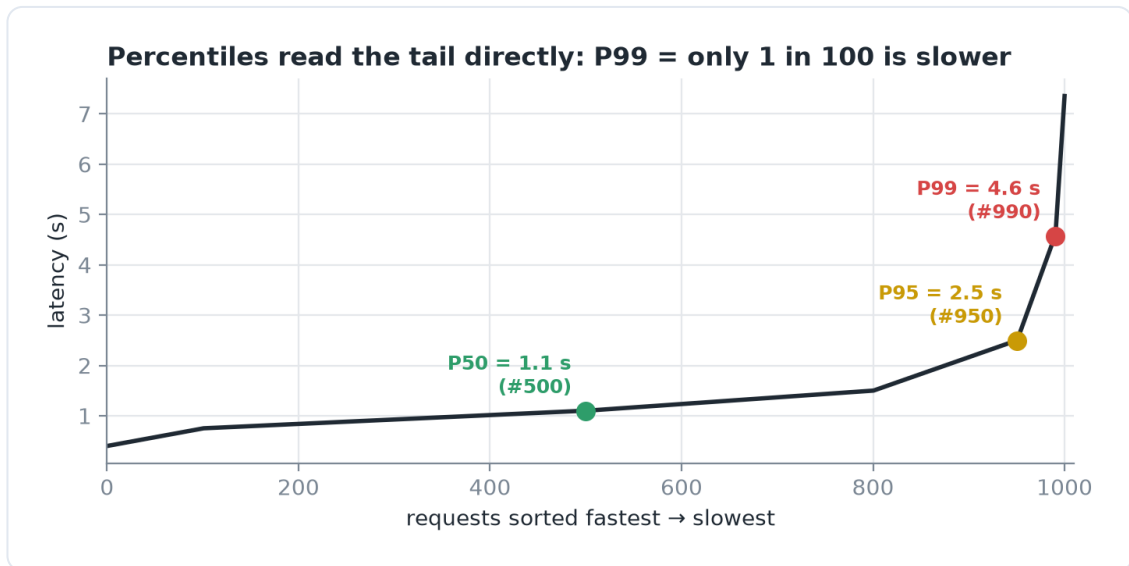
1,000 requests (illustrative): most near 1 s, but queueing and long replies stretch a tail to the right.

Summary	How	Here	Use
Mean	Sum ÷ count	1.3 s (dragged right by the tail)	Describes almost nobody when the tail is long
Median (P50)	Middle request: half faster, half slower	1.1 s	The typical request

The tail, the slowest requests, is exactly what users complain about.

4. Percentiles describe the tail

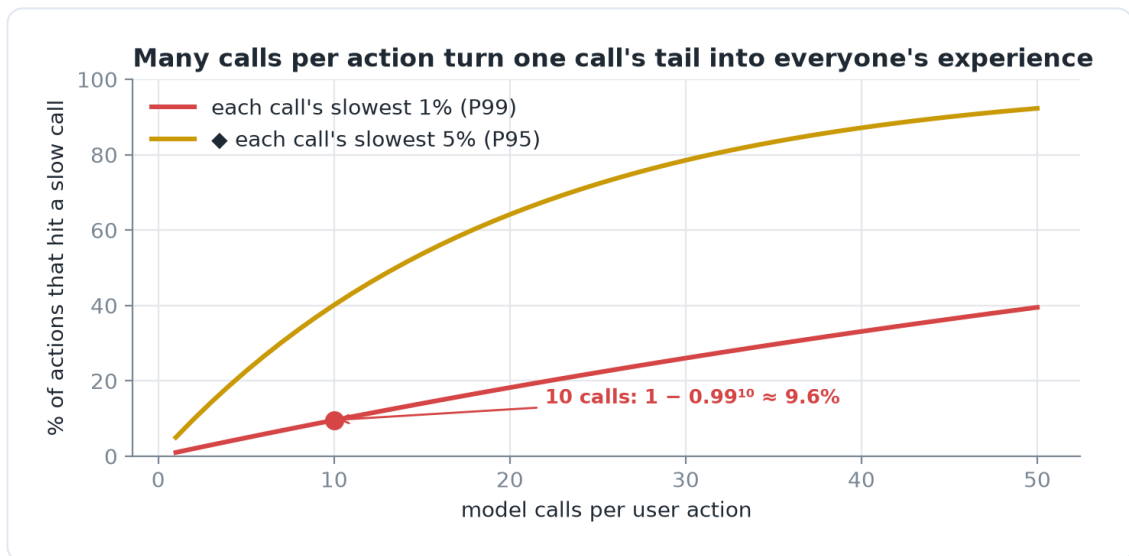
Sort the 1,000 latencies. **P50** is the value at position 500, **P95** at 950 (95% are faster), **P99** at 990 (only 1 in 100 is slower).



Here P50 = 1.1 s, P95 = 2.5 s, P99 = 4.6 s.

Why care about 1 in 100? Real pages and agents make **many calls** per user action:

$$P(\text{at least one of 10 calls in the slowest 1\%}) = 1 - 0.99^{10} \approx 9.6\%$$



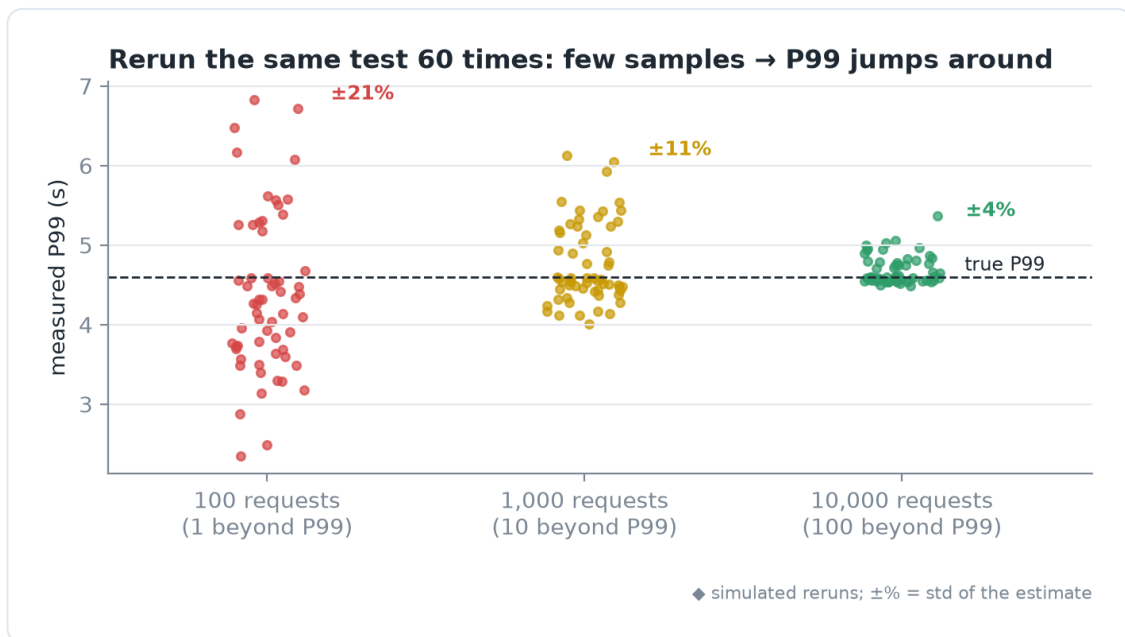
The more calls per action, the more often someone hits a slow one.

KEY IDEA

The tail of each call becomes the typical experience of the whole task. That's why serving targets are percentiles, e.g. **P99 time to first token < 2 s**.

5. How many samples does a percentile need?

With only **100 requests**, P99 is the value at position 99, decided by the **single slowest request**. Rerun the test and that one request changes, so P99 jumps around.



Simulated reruns: P99 from 100 requests swings widely; from 10,000 it barely moves.

Rule of thumb: about **100 samples beyond the percentile** you report gives an estimate steady to about $1/\sqrt{100} = 10\%$.

Report	Fraction beyond	Requests needed
P50 (median)	50%	200
P95	5%	2,000
P99	1%	10,000

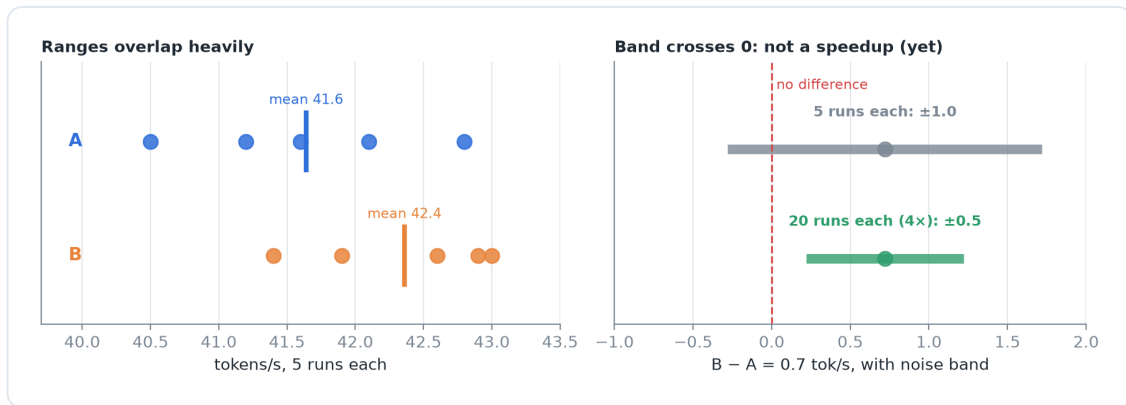
INTUITION

Whenever you see a P99, ask: **out of how many requests?**

6. Real speedup, or noise?

Back to 41.6 vs 42.4 tok/s. The honest move: **run each setting several times**. Five runs each: A lands from ~ 40.5 to nearly 43, B from ~ 41.5 to 43. The ranges **overlap heavily**.

- **Standard deviation** (spread of runs): $\sigma \approx 0.8$ tok/s.
- **Standard error** of an average shrinks with $\sqrt{n}$: $SE = \sigma/\sqrt{n} = 0.8/\sqrt{5} \approx 0.4$.
- For the **difference of the two averages**, the noise band is about ± 1 tok/s. The observed difference is only **0.7**, inside the noise.



With 5 runs each the band crosses zero: can't claim B is faster. 4 $\times$ the runs halves the band; if the 0.7 held up, it would clear zero.

KEY IDEA

Before calling something a speedup, compare it to the **run-to-run noise**.

7. Recap

Lesson 06 on one card

OUTPUT IS A DRAW

the model gives a distribution
each token is a fresh random draw
so text and lengths vary

REPORT THE TAIL

latency is skewed: mean $\neq$ typical
report P50 / P95 / P99
many calls amplify the tail

SAY HOW MANY

~ 100 samples beyond the percentile
P50: 200 · P95: 2k · P99: 10k
→ about $\pm 10\%$

BEAT THE NOISE

$SE = \sigma / \sqrt{n}$
a difference counts only if
it's bigger than run-to-run noise

Interview cheat sheet

Revise from these pages alone. ♦ = my addition beyond the video; everything else is from the lesson. Builds on 01–05.

Say it in 30 seconds

Each generated token is a random draw from the model's distribution, so reply text and length vary and latency becomes a long-tailed distribution. Report P50, P95 and P99, not the mean, and state the sample count: you need about 100 samples beyond a percentile for ~10% precision. Many calls per task amplify the tail. Treat a speedup as real only if it exceeds run-to-run noise, measured with the standard error $\sigma/\sqrt{n}$.

Only numbers worth memorizing

- **Samples for ±10%:** P50 → 200, P95 → 2,000, P99 → 10,000 requests. **10 calls at P99:** ~9.6% of tasks hit the tail.

Derive, don't memorize

1. Tail amplification: chance at least one of k calls lands in the slowest fraction q

$$P = 1 - (1 - q)^k = 1 - 0.99^{10} \approx 9.6\%$$

INTUITION

For small q , $P \approx k \cdot q$ ♦. An agent making 50 calls hits a P99-slow call ~40% of the time: set per-call targets at a high percentile.

2. Samples needed: ~100 beyond the percentile, error $\approx 1/\sqrt{100}$

$$n \approx \frac{100}{1 - p} \quad \text{P99: } \frac{100}{0.01} = 10000 \quad \text{error} \approx \frac{1}{\sqrt{100}} = 10\%$$

3. Noise band: standard error shrinks with $\sqrt{n}$

$$\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{5}} \approx 0.4 \quad \text{band}_{\text{diff}} \approx 2 \times \sqrt{2} \times \text{SE} \approx \pm 1.0$$

INTUITION

♦ $\sqrt{2}$ because both averages are noisy; $\times 2$ for ~95% confidence. $4 \times$ the runs halves the band. $0.7 < 1.0$: not proven.

Mean vs median vs percentiles

	Mean	Median (P50)	P95 / P99
Tells you	Average cost (capacity math) ♦	The typical request	The tail users complain about
Long tail effect	Dragged right (1.3 vs 1.1 s)	Unaffected	Is the tail
Samples needed	Few	~200	~2,000 / ~10,000

Symptom → diagnosis

Symptom	Diagnosis
Mean latency fine, users complain	Long tail: look at P95/P99
P99 jumps between identical reruns	Too few samples beyond P99
"2% faster" from one run each	Compare to run-to-run noise first
Same prompt, very different request times	Sampled reply lengths differ

Rapid-fire Q&A

Question	Crisp answer
Why does the same prompt give different outputs?	Each token is a random draw from a distribution.
Why are SLOs written as percentiles?	The tail is what users feel, and multi-call tasks amplify it.
How do you know a speedup is real?	Repeat runs; the difference must exceed the noise band.
◆ How do temperature / top-p relate?	They reshape or truncate the distribution before the draw.

TRAP

- Reporting only the mean for a long-tailed latency distribution.
- Quoting P99 from a few hundred requests without saying so.